Molecules and compounds

☞ **1** Use the words in the word bank to complete the following passage.

All matter is made up of a_____. Atoms are rarely found on their own. They are often found joined together, forming m_____.

In molecules, the atoms that are chemically joined together may all be the same or they may be d_____. We can have molecules of an e_____ – for example, hydrogen (H_2) or oxygen (O_2) – or we can have molecules of a c_____. Compounds are substances that are made up of t_____ or more elements chemically joined together. For example, w_____ (H_2O) and ammonia (NH_3) are both compounds. The s_____ H_2O and NH_3 are the formulas for water and ammonia. A chemical formula tells us which t_____ of atoms are present and how m_____ atoms of each type there are per molecule.

Word bank
symbols
atoms
different
molecules
two
water
type
element
many
compound

2 Classify each of the following as either a molecule of an element or a molecule of a compound.

P_4 _____ I_2 _____

HCl _____ CO _____

CH_4 _____ S_8 _____

3 Complete the following information. The first one has been done for you.

Substance	Formula	Each molecule contains
water	H_2O	2 atoms of hydrogen, 1 atom of oxygen
magnesium oxide	MgO	
salt (sodium chloride)	NaCl	
carbon dioxide	CO_2	
sugar	$C_{12}H_{22}O_{11}$	
oxygen gas	O_2	
ammonia gas	NH_3	

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